

AWS · Azure · GCP

Pick right, and you focus on product. Pick wrong, and the company dies on compute. DPDP adds residency rules; customer preferences shape choices; team skills decide velocity.

What would your startup's cloud bill be if 50k users queried your app daily?

L3

WHAT'S INSIDE

- 01 3 clouds side-by-side
- 02 DPDP + residency
- 03 Core AI services you actually use
- 04 Typical Indian AI startup — ₹ 2.45...
- 05 Picking framework

WHY THIS LESSON MATTERS

30–50% of an Indian AI startup's tech budget is cloud.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Compare AWS, Azure, GCP for ML.

2

Core AI services of each.

3

Deploy with SageMaker / Azure ML / Vertex.

4

Choose cloud by skills + residency + customer.

5

Estimate monthly bill for Indian AI startup.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

3 clouds side-by-side

Key concept of this lesson.

02

DPDP + residency

The 2023 Indian law that shapes architecture

03

Core AI services you...

Mapping across vendors

04

Typical Indian AI startup —...

LLM costs DOMINATE. Optimise there FIRST before touching infra.

01

PART 1 OF 4

3 clouds side-by-side

Key concept of this lesson.

CONCEPT 1 OF 4

3 clouds side-by-side

A core idea you'll use throughout this lesson.

DEFINITION

3 clouds side-by-side

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

DPDP + residency

Significant personal data MUST stay in India.

Always deploy in Mumbai region by default.

Verify LLM API routing passes through India.

Encryption at rest + in transit — always.

DEFAULT

ap-south-1

Mumbai first; other regions with legal sign-off.

KEY POINTS

- Significant personal data MUST stay in India
- Always deploy in Mumbai region by default
- Verify LLM API routing passes through India
- Encryption at rest + in transit — always

CONCEPT 3 OF 4

Core AI services you actually use

Notebooks: SageMaker Studio / Azure ML / Vertex Workbench.

Training: SageMaker Training / Azure ML Jobs / Vertex Training.

Registry: 3 equivalents.

Endpoints: 3 equivalents.

LLMs: Bedrock / Azure OpenAI / Vertex Model Garden.

ONE-TO-ONE

Services map

Pick by team skill + pricing.

DEFINITION

Core AI services you actually use

Mapping across vendors

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Typical Indian AI startup — ₹ 2.45 lakh/month

LLM costs DOMINATE. Optimise there FIRST before touching infra.

KEY POINTS

- LLM costs DOMINATE. Optimise there FIRST before touching infra
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Price your W5-P1

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which of the 3 clouds would YOUR customers prefer?

Q2

Where would DPDP bite if you chose a non-Mumbai region?

Q3

Which lever (compute vs LLM) would you attack first?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

SageMaker equivalent on GCP:

A

Azure ML

B

Vertex AI

C

MLflow

D

DVC

SageMaker equivalent on GCP:



CORRECT ANSWER

Vertex AI

WHY

Vertex AI

Default region for Indian personal data:

A

us-east-1

B

ap-south-1

C

eu-west-1

D

global

Default region for Indian personal data:



CORRECT ANSWER

ap-south-1

WHY

Mumbai — DPDP residency

Biggest cost line for a typical Indian AI startup:

A

Infra

B

LLM API

C

S3

D

DB

Biggest cost line for a typical Indian AI startup:



CORRECT ANSWER

LLM API

WHY

Often 50-70% of cloud bill

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

"And your customers already trust."

WHAT TO NOTICE

1

3 clouds side-by-side

2

DPDP + residency

3

Core AI services you...

4

Typical Indian AI startup —...

Cloud cost estimate for W5-P1

Price your knowledge assistant across AWS, Azure, GCP at 5k queries/day.

DELIVERABLES

- Compute sizing.
- LLM API cost.
- Storage + networking.
- Monthly total × 3 clouds.
- Recommendation + why.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Spreadsheet + 2-sentence recommendation.

ONE THING TO REMEMBER

“

"And your customers already trust."

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — Monitoring + drift detection.